

Bus Management System

Presented by



Meherab Alam Rion
253-15-192



Nova Akther
253-15-716



Rifa Shanzida
253-15-805



Tamim Ahmed
253-15-761



Azizul Hakim Shuvo
253-15-849

Submitted to
Jamiul Haque Jami



Introduction

What is Bus Management System?

A Bus Management System is a software that helps manage buses and passengers efficiently.

Main Purpose

- Reduce manual work
- Manage buses efficiently
- Organize passenger information
- Improve seat management
- Apply Data Structure concepts in a real-life system



Meherab Alam Rion
253-15-192



Problem Statement

01

Existing Problems

- Manual ticket booking
- Difficult seat management
- No waiting list
- Slow searching process
- Difficult to manage buses
- Feedback is not organized

02

Our Solution

We developed a Bus Management System using Data Structures to make the process faster and easier

Meherab Alam Rion
253-15-192

Project objectives



Obtain Objectives

- Search available bus routes
- Book seats easily
- Cancel booked seats
- Display seat availability
- Collect passenger feedback
- Manage buses through the admin panel
- Maintain a waiting queue when seats are full
- Improve data organization and efficiency



Meherab Alam Rion
253-15-192



System Structure

2030

BUS MANAGEMENT SYSTEM

Structure:

Passenger Panel : Search Route, Book Seat, Cancel Seat,
| | | | | Seat Map, Feedback

Admin Panel : Add Bus, Delete Bus, Update Bus,
| | | | | View All Buses, View Feedback,
View Waiting Queue

Bus Linked List

| | |
Seat Array + Feedback Array + Waiting Queue (FIFO)

===== */



Nova Akther
253-15-716

Data Structures Used

01 Data Structures Implemented

Data Structure Purpose

Linked List :Store and manage bus information

Array :Store seat status of each bus

Array:Store passenger feedback

Queue (FIFO):Manage waiting passengers



Nova Akther
253-15-716

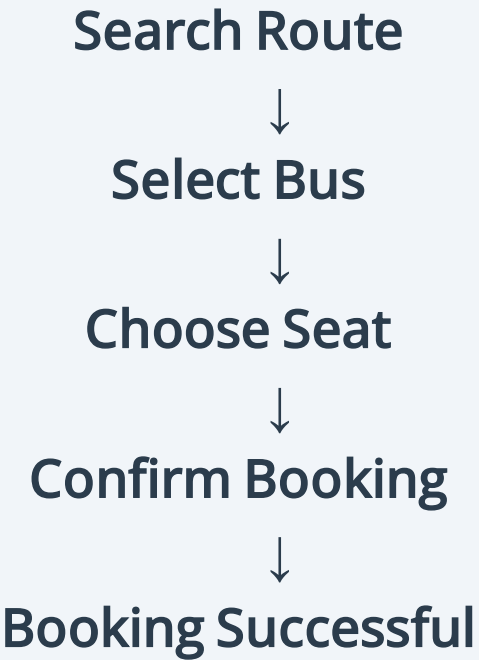


Passenger Panel

The Passenger Panel provides the following services:

- Search Bus Route
 - Book Seat
- Cancel Booking
- View Seat Map
- Submit Feedback

Passenger Workflow





2030

Admin Panel

Admin Features

Admin controls the whole system.

- Add Bus
- Delete Bus
- Update Bus
- View All Buses
- View Feedback
- View Waiting Queue

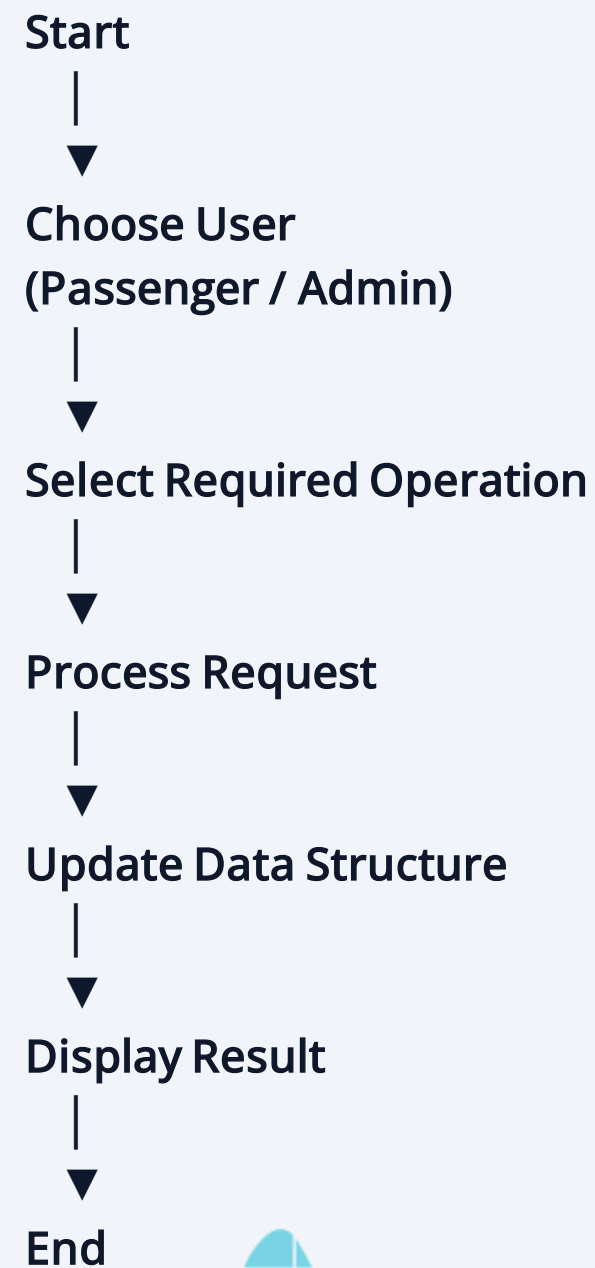
Responsibilities

- Maintain bus records
- Monitor passenger activities
- Keep data updated
- Improve system management

Rifa Shanzida
253-15-805



Working Process of the System



Explanation

Every operation performed by the Passenger or Admin updates the corresponding data structure, ensuring accurate and efficient data management.

Rifa Shanzida
253-15-805



2030

Project Algorithms

01

Linked List

- Insert Bus
- Delete Bus
- Traverse Bus

02

Array

- Seat Booking
- Seat Cancellation

04

Queue

- Enqueue Waiting Passenger
- Dequeue Passenger

Tamim Ahmed
253-15-761



Advantages of Our System

Advantages

- Fast searching
- Easy booking
- Organized data
- Easy bus management
- Better seat management
- Waiting list support
- User-friendly

Outcome

- ✓ Saves Time
- ✓ Reduces Manual Work
- ✓ Improves Management Efficiency



Tamim Ahmed
253-15-761



Future Improvements

Future Version may include

- Online Payment
- User Login System
- QR Ticket
- Email Notification
- Mobile App
- Database Integration
- GPS Bus Tracking

Goal

To transform this academic project into a complete real-world Bus Management System.

Note: The following features are proposed for future development and are not part of the current implementation.



Azizul Hakim Shuvo
253-15-849



Team Contribution

01

Member	Responsibility
1) Meherab	Passenger Module
2) Nova	Admin Module
3) Rifa	Linked List
4) Tamim	Queue & Array
5) Shovo	Testing Documentation

02

Our Solution

We developed a Bus Management System using Data Structures to make the process faster and easier

Azizul Hakim Shuvo
253-15-849



Conclusion

Project Summary

The Bus Management System is a simple and efficient application developed using Data Structure concepts.

Data Structures Used

- Linked List → Bus Information
- Array → Seat Management
- Array → Feedback Management
- Queue (FIFO) → Waiting List



Conclusion

Our project demonstrates how Data Structures can solve real-life management problems efficiently. It is easy to use, well organized, and can be expanded into a complete software system in the future.

Azizul Hakim Shuvo
253-15-849



Thank
you!

